

Highlights

From Augmentation to Substitution: Delegation Habits, Skill Atrophy, and the Limits of Trust Calibration in Human-AI Collaboration

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- Trust calibration cannot explain habitual, embedded agentic-AI use in knowledge work
- Delegation habits develop in four stages, culminating in skill atrophy
- Habitual over-delegation erodes the very skills needed when AI fails
- Aviation and human-robot collaboration evidence grounds claims for cognitive AI
- Validated instruments and system-log indicators make delegation habits measurable

From Augmentation to Substitution: Delegation Habits, Skill Atrophy, and the Limits of Trust Calibration in Human-AI Collaboration

Samuele Dell'Oca^{a,*}, Valentina Rotondi^b, Vincenzo Cutrona^a, Giuseppe Landolfi^a, Andrea Bettoni^a, Luca Canetta^a

^a*University of Applied Sciences and Arts of Southern Switzerland, Department of Innovative Technologies, Lugano, Switzerland*

^b*University of Applied Sciences and Arts of Southern Switzerland, Department of Business Economics, Health and Social Care, Manno, Switzerland*

Abstract

The accelerating deployment of agentic artificial intelligence in industrial work environments has exposed a gap between what the dominant theoretical lens, trust calibration, was built to explain and the behavioral reality of sustained human-AI interaction. Trust calibration theory accounts for episodic reliance decisions informed by an appraisal of system reliability; whereas in repeated, embedded AI use, reliance behavior progressively decouples from that appraisal. The result is automatized delegation routines that operate below the threshold of deliberate decision-making, and measurable degradation of the cognitive capabilities those routines replace. This critical review synthesizes four bodies of literature (human-AI trust research, habit psychology, automation complacency from aviation and Human-Robot Collaboration, and cognitive offloading theory) into the *Delegation Habit Framework* (DHF), a conceptual model that reconceptualizes human-AI reliance as a habit-formation process unfolding in four progressive stages: formation, crystallization, divergence, and skill atrophy. The framework repositions industrial knowledge work as the next frontier of a dynamic already documented in physical human-machine teaming, where automation complacency and capability degradation have long been recognized as safety-critical outcomes. We operationalize the DHF through adapted self-report measures

*Corresponding author

Email address: `samuele.delloca@supsi.ch` (Samuele Dell'Oca)

of behavioral automaticity, experience sampling protocols, skill retention assessments, and behavioral indicators derivable from industrial information systems, and articulate a structured research agenda with eight prioritized research questions. Implications for system design, organizational AI governance, and regulatory compliance are discussed. The dividing line between augmentation and substitution, we argue, is behavioral: AI augments capability only while delegation remains a deliberate choice.

Keywords: human-AI collaboration, delegation habits, skill atrophy, trust calibration, cognitive offloading, Operator 5.0

1. Introduction

Artificial Intelligence (AI) is no longer an advisory tool that workers consult at discrete decision points: AI systems increasingly operate as persistent, embedded agents that draft communications, synthesize information, monitor processes, and execute multi-step tasks with minimal human initiation [1]. The transition is accelerating in industry: a majority of knowledge workers now interact with AI multiple times daily, across tasks from process monitoring and quality inspection to engineering design support and supply chain decision-making [2]. Whether workers will adopt AI is no longer in question: they already have. The pressing question is what sustained, embedded AI use does to the people who rely on it.

The dominant framework for answering this question is *trust calibration* [3], a paradigm whose status as the field’s organizing principle has been confirmed by multiple independent systematic reviews [4, 5, 6]. It holds that appropriate human reliance on automation depends on the alignment between a worker’s subjective trust and the system’s objective reliability: well-calibrated trust yields appropriate reliance, while miscalibrated trust — too high (over-reliance) or too low (under-reliance) — yields dysfunctional reliance. This framework has generated decades of valuable research, identifying the determinants of trust [7, 8] and its dynamics under failure [9]. Yet it was designed to explain a specific human-automation interaction: one in which a worker makes a deliberate, traceable decision to accept or override an automated recommendation. It was built for the cockpit and the control room, not for the always-on AI assistant.

This paper argues that the mismatch is structural, not incidental. What begins as a deliberate reliance decision progressively becomes a context-cued

routine (a habit) operating largely below the threshold of explicit trust appraisal [10], a trajectory documented empirically as a shift from initial algorithm aversion to functional AI dependence [11]. Once delegation is habitual, workers no longer consciously re-evaluate AI reliability at each delegation; they respond to contextual triggers reinforced by prior use. The result is a systematic drift toward over-delegation whose consequences extend beyond any single decision: the cognitive capabilities that repeated delegation replaces gradually deteriorate from disuse, a phenomenon named *skill atrophy* [12, 13]. For knowledge workers and the organizations that employ them, the critical question is whether sustained AI use erodes the very capabilities required to manage, supervise, and recover from system failures.

This dynamic has precedent in physical human-machine systems, where capability degradation is a known outcome of successful automation. In aviation, sustained reliance on automated flight systems reduced manual flying proficiency and diminished situation awareness in failure conditions [14, 15]. In Human-Robot Collaboration (HRC), repeated reliance on robotic assistance has been associated with declines in operator task-specific expertise [16, 17]. The transition to cognitive human–AI collaboration does not eliminate this dynamic but extends it to industrial knowledge work, where the capabilities at stake include analytical reasoning, professional judgment, and domain expertise.

This paper is a critical review with a conceptual contribution and a research agenda: it contains no new empirical data, and its claims are framed as theoretically grounded propositions to be tested, not established findings. Within that genre, it makes three contributions. First, drawing on a cross-field synthesis of four bodies of empirical literature (selection approach in Section 3.1), it identifies the scope conditions under which trust calibration theory is insufficient for agentic AI use and proposes habit formation as a complementary mechanism for the repeated-use context. Second, it introduces the Delegation Habit Framework (DHF), a conceptual model that traces AI delegation through four progressive stages — *Formation*, *Crystallization*, *Divergence*, and *Skill Atrophy* — describing how delegation routines become automatic, drift toward over-delegation, and measurably degrade the cognitive capabilities they replace; each stage is grounded in converging findings from the four literatures. Third, it operationalizes the DHF with validated instruments, study designs, and indicators derivable from industrial information systems, grounds it in an illustrative industrial scenario, and articulates a structured research agenda connecting the framework to

the human-AI symbiosis challenge facing industrial organizations.

Throughout the paper, we refer to “AI system” broadly: the term covers conversational assistants used interactively, models invoked through programmatic interfaces, and AI functionality embedded directly within applications. These forms share what matters here, the delegation of cognitive work to a machine; they differ in how visible the delegation moment is and in how much of the input the worker shapes (for instance, through prompting), differences the DHF treats as moderators of habit formation rather than as boundaries of its scope.

The paper is organized as follows. Section 2 examines the limits of trust calibration theory in the agentic era. Section 3 sets out the literature selection approach and synthesizes the four bodies of literature on which the DHF is built. Section 4 presents the framework itself. Section 5 develops the bridge from physical to cognitive human-machine teaming. Section 6 provides the operationalization. Section 7 proposes the research agenda. Section 8 discusses implications and limitations, and Section 9 concludes.

2. The Trust Calibration Paradigm and Its Limits in the Agentic Era

2.1. What trust calibration theory is and what it was built for

In industrial AI applications—from predictive maintenance dashboards to quality control systems—trust calibration has long been the dominant lens on human–automation interaction. The concept originates in [3], which defined trust in automation as “the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability.” Two clarifications in that account matter here. First, [3] is explicit that trust is an *attitude* whereas reliance is a *behavior*; trust is one input to the reliance decision, alongside workload, self-confidence, time pressure, and effort of engagement. Second, the framework’s practical leverage rests on the linkage between the two: appropriate reliance follows from appropriate trust because the trust attitude is assumed to remain an active input to each reliance decision. [18] elaborated a typology of reliance failures — misuse (over-reliance), disuse (under-reliance), and abuse (design-induced miscalibration) — that still structures research in this domain.

Historically, trust calibration theory was developed in aviation, process control, and military systems, domains characterized by discrete, high-stakes human–automation decisions and explicit opportunities for monitoring and

override [14, 15]. In these contexts, the assumption that reliance reflects a deliberate, recurrent trust appraisal is appropriate. More recent work has extended the framework to account for dynamic trust formation and co-evolution between trust and system performance [19], as well as to human–AI decision-making contexts [20]. These extensions enrich the framework, but they retain its linkage assumption: that reliance behavior remains informed by the trust attitude, even when that attitude has been updated.

2.2. *Three boundary conditions in the agentic context*

In sustained industrial AI use this assumption becomes strained: agentic AI introduces conditions under which reliance behavior departs from the scope of episodic trust calibration, and delegation to AI is already known to produce unexpected performance effects [21]. We identify three boundary conditions that help explain why.

Repetition and habituation. Trust calibration’s leverage on behavior presupposes that the trust attitude informs each reliance decision. This linkage weakens when workers repeatedly delegate the same task types—report summarization, anomaly detection, decision support—dozens of times per day. Over repeated exposure, reliance behavior becomes increasingly stable and less sensitive to situational variation [7]; habit theory holds that repeated behaviors in stable contexts become automatic, triggered by cues rather than deliberation [10]; and automation bias evidence shows verification declining with repeated exposure [22, 23]. Workers may continue delegating even when they intend otherwise. At the limit, the trust appraisal is not updated—it is bypassed.

Embeddedness and invisibility. Trust calibration assumes a visible delegation decision: the user knows when automation acts and can decide whether to rely on it. Agentic AI blurs or removes this boundary: outputs arrive pre-populated, pre-filtered, or integrated into workflows without a clear moment of delegation, so users encounter results rather than decisions. The absence of a visible handoff removes the trigger for explicit trust evaluation, a cognitive extension of the “irony of automation” [12]: the more effectively systems perform, the fewer opportunities users have to monitor or intervene.

Temporal scale and capability consequences. Trust calibration is episode-focused: it evaluates whether reliance is appropriate in a given instance, not how repeated reliance affects human capability over time. Where AI is used continuously, the relevant question shifts from immediate decision

quality to long-term capability maintenance, and skill atrophy, the degradation of cognitive capabilities through disuse, lies outside the traditional scope of trust-based accounts.

2.3. An insufficient construct for the agentic era

Taken together, these boundary conditions suggest that trust calibration remains highly effective for episodic human–automation interaction but has limited explanatory power in sustained, embedded AI use. The argument is not that trust calibration is wrong, but that its scope is bounded: it explains the trust attitude and the reliance decisions that attitude informs, while the post-calibration regime, in which reliance decouples from the trust attitude and persists as a cue-triggered routine, requires a behavioral mechanism the framework does not supply. The DHF targets precisely that reliance side of the distinction. Parallel work points the same way: even well-calibrated trust is insufficient for optimal decision-making without accurate self-monitoring [24]. Habit theory provides the complementary lens: it explains how repeated behaviors become automatized, why they persist independently of conscious evaluation, and how sustained AI delegation may reshape both behavior and capability over time. For industrial organizations, managing AI adoption is therefore not only a matter of calibrating trust, but of monitoring emerging behavioral routines and maintaining critical human capabilities. The next section synthesizes the relevant literature.

3. Four Theoretical Pillars: A Cross-Field Synthesis

The DHF integrates four bodies of literature, each contributing an explanatory element the others cannot provide alone. This section develops each in turn before an integrative synthesis, and each pillar closes with a concrete illustration of how its core dynamic manifests in current AI-supported knowledge work.

3.1. Scope and literature selection approach

This is a critical review rather than a systematic one: it aims not at exhaustive coverage but at theoretical adequacy — the minimal set of literatures required to explain two phenomena each field captures only partially: the automatization of AI reliance, and the capability consequences of sustained delegation. The four pillars follow deductively from these two

explananda: trust calibration is the incumbent account whose boundary conditions define the problem (Section 2); habit psychology supplies the automatization mechanism; automation complacency research supplies the closest empirical precedent in physical human-machine systems; and cognitive offloading theory links externalized cognition to capability change. Within each pillar, sources were prioritized consistently: foundational theoretical statements, then systematic reviews and meta-analyses, then recent (2024–2026) AI-specific empirical work, located through database and citation searching complemented by AI-assisted discovery, with every source verified by the authors. Studies were included when they spoke directly to one of the four stages or to the trust–reliance linkage; adjacent literatures (e.g., technology acceptance) were consulted but not synthesized where their contribution duplicated a pillar. The same criteria govern the stage-level evidence mapping in Section 3.6.

3.2. Human-AI trust research: the problem well-described, the mechanism absent

The human-AI trust literature has established, with considerable rigor, the factors shaping initial trust formation and the consequences of miscalibrated trust. Independent systematic reviews confirm trust calibration as the field’s central organizing construct: across a database-screened analysis of 1,697 articles [4], a survey of 96 high-quality publications that finds trust “the central variable to explain both under-trust and over-trust” [5], and a 30-year review tracing the framework’s unbroken influence from automation to current AI contexts [6]. Dispositional, situational, and learned trust all influence reliance decisions [7]. Automation bias (the tendency to accept automated recommendations without sufficient verification [22]) is documented from aviation to medical diagnosis to legal judgment, and is most pronounced under cognitive load and time pressure.

What this literature does not provide is a mechanism for what happens *after* trust has stabilized. The practical leverage of trust calibration models rests on the trust attitude remaining an active input to each reliance decision, and the evidence suggests this linkage erodes: trust in familiar automation becomes largely dispositional and invariant over time [7], and automation bias persists even when explanations are provided [23]. [21] showed that productive human-AI teaming requires active human contribution — yet when delegation becomes the default, workers progressively cede that contribution. [25] showed that contextual information moderates delegation, implying that

in low-information or routine conditions (exactly when habit cues dominate), delegation is most likely to become automatic. [26] associated generative AI use with self-reported reductions in critical thinking effort, and [27] synthesized sixty-two studies documenting reduced independent diagnostic capability across clinical domains. The trust literature treats persistent over-reliance as a calibration failure and addresses it through transparency and explainability; the habit literature explains why this fails post-calibration: once reliance is automatic, it is by definition resistant to information-based interventions. Nor does the trust literature track capability: its outcome variable is the quality of individual reliance decisions, not what sustained reliance does to the skills of the person making them. *In AI delegation terms*: a process engineer who, after months of reliable experience, accepts an AI-generated anomaly classification without opening the underlying sensor data exhibits exactly this post-calibration behavior. Their expressed trust may not have changed in months; what has changed is that the trust appraisal no longer mediates acceptance — and no transparency dashboard can re-engage an evaluation that is no longer performed.

3.3. *Habit psychology: the mechanism the trust literature needs*

Habit psychology, as developed by Wood, Neal, and colleagues [10, 28], defines habits as context-response associations acquired by repeating a behavior in a stable context: when a behavior recurs in the presence of the same cues, the cue-behavior association is gradually encoded in memory, until the cue activates the behavior automatically, without deliberate intention. [29] demonstrated that habits persist even when actors consciously intend otherwise and when the behavior is no longer instrumentally rational: under strong cue presence, the “pull of the past” exceeds the pull of current intention.

Habit formation follows a characteristic trajectory. [30] found that automaticity (the experience of performing a behavior without deliberate initiation) grows asymptotically to a plateau, with time-to-plateau varying widely (18 to 254 days in their sample) but a consistent curve shape: early rapid growth, then stabilization. Once at plateau, a behavior is maintained by contextual cues rather than motivation or deliberation, and context removal is the most reliable route to disruption [10].

The Self-Report Behavioral Automaticity Index (SRBAI) [31], derived from the Self-Report Habit Index [32], is a validated four-item instrument for the automaticity component of habit strength, measured independently of behavior frequency. This distinction matters for AI delegation: a worker

who delegates daily but experiences each delegation as a deliberate choice is calibrating; one who delegates at the same frequency without conscious initiation is habituating. The SRBAI has not yet been applied to AI delegation in published research; the DHF’s empirical agenda (Section 6) proposes exactly this application.

In AI delegation terms: the habit construct maps directly onto knowledge work. The *cue* is a recurrent task trigger: an incoming nonconformity report, a request to evaluate supplier options, a blank document; the *response* is opening the AI assistant and pasting the material in; the *reinforcement* is a usable draft within seconds at near-zero effort. Because these digital cues recur with exceptional stability (the same notification, interface, and one-click entry point, many times daily), the association can consolidate faster than for the offline behaviors habit research has typically studied. The theory also specifies that disruption requires modifying the cue environment itself [10], not better information or stronger intentions (e.g., removing the assistant from the default workflow for selected task classes, a design lever developed in Section 8).

3.4. Automation complacency: the empirical precedent from physical systems

The automation complacency literature, originating in aviation safety research, established the foundational empirical case that high automation performance degrades human supervisory capability [14]. Complacency is reduced vigilance and monitoring during prolonged smooth, reliable automation: the operator who has never had to intervene loses both the skill and the inclination to do so. Situation awareness theory [15] supplies the mechanism: well-performing automation removes the need for active monitoring, and operators who stop monitoring progressively lose the mental model of system state needed to identify failures and intervene.

Decades of accident investigation have catalogued the consequences. The 2009 Air France 447 accident (the autopilot disconnected following pitot tube failure and the crew could not regain manual control at altitude [33]) and the 2013 Asiana Airlines 214 accident (the crew failed to manage airspeed manually after deactivating the autothrottle [34]) both document degraded manual proficiency under sustained automation dependency: the automation had performed reliably so long that manual skills had eroded. Beyond trust calibration, these are failures of capability maintenance that trust-focused accounts do not address — precisely the dynamic Bainbridge [12] anticipated. Section 5 extends this precedent to cognitive human-machine teaming.

The HRC literature has documented an analogous dynamic in physical collaborative work, raising systematic concern about operator skill retention when robot assistance is withdrawn [17]. The Operator 4.0 framework [35] anticipated this in its emphasis on human augmentation (technology should amplify human capabilities, not substitute for them) and the Operator 5.0 trajectory [16] extends the concern to cognitive augmentation, asking what happens to human cognitive capabilities in work systems where AI handles an increasing share of information processing. What this literature documents is the outcome, degraded supervision under reliable automation in physical settings; it does not supply the invisible, persistent habit mechanism that produces it, nor its extension to cognitive work.

In AI delegation terms: the supervisory role does not disappear in cognitive AI use — it relocates. Depending on the system, the worker may also shape the input (framing the prompt, selecting the context the system sees), so involvement is not confined to the receiving end; the supervision construct, however, concerns the downstream side: *output verification before acceptance*. The knowledge worker who reviews a Large Language Model (LLM)-drafted summary or checks an AI-proposed root-cause hypothesis against the maintenance history occupies the same structural position as the pilot monitoring an autopilot. The complacency analog is thus directly observable in everyday AI work (accepting outputs unread, copy-pasting generated text into deliverables) and declining verification engagement is already documented in conversational AI interaction [36, 26]. The cognitive case adds an aggravating factor: when the user frames the AI as a peer rather than as automation to be supervised, the supervisory framing is not eroded over time — it is never instantiated at all. The monitoring role that aviation formalized in procedures and checklists has, in most AI-supported knowledge work, no institutional expression whatsoever.

3.5. Cognitive offloading theory: the mechanism of skill atrophy

Cognitive offloading theory [13] describes how individuals externalize cognitive operations (memory storage, calculation, inference, decision-making) to artifacts and agents in the environment. Offloading is adaptive in principle: distributing cognitive work across internal and external resources lets individuals perform tasks exceeding their unassisted capacity. The concern is not offloading per se but its cumulative effect on the internal resources being bypassed.

The mechanism is the use-it-or-lose-it principle: capabilities not regularly exercised degrade over time, and the principle extends beyond memory and motor skills to professional cognitive capacities. [37] demonstrated that the mere availability of external information retrieval reorganizes memory encoding — people encode where to access information rather than the information itself. [38] established automation-induced skill degradation as a replicated phenomenon across aviation, medicine, and process control, and [39] associated AI tool reliance with reduced critical thinking performance, consistent with mediation through cognitive offloading. [13] note that the strategic value of offloading depends on the external resource remaining reliably available: a worker who has offloaded critical reasoning to an AI agent has bet that the agent will always be functional, and the failure modes of that bet (outage, hallucination, domain transfer, regulatory restriction) are precisely the conditions under which the degraded internal capability becomes safety-relevant.

A useful reframing: cognitive skill atrophy from sustained delegation is the default expectation, not a novel prediction. Calculators have been linked to reduced mental arithmetic fluency, GPS navigation to degraded spatial memory, internet search to shallower associative encoding [37]; the effect of computer-aided detection on radiologists’ diagnostic sensitivity is a professional-domain case [38]. Across every cognitive tool that has systematically reduced the need for a specific human cognitive operation, measurable degradation has consistently followed unless deliberate countermeasures were in place. The burden of proof lies with those who would argue AI delegation will be the first exception.

Recent AI-specific work provides converging evidence: accelerated skill decay without performers’ awareness [40]; reduced independent performance immediately after AI-assisted use [41]; co-occurring declines in verification confidence and task accuracy in a three-wave longitudinal study [42]; the gradual, “asymptomatic” judgment deterioration that [43] term *intuition rust* — invisible precisely when most advanced; and the argument that AI systems create “capacity-hostile environments” making atrophy self-reinforcing at organizational scale [44].

The offloading framework also distinguishes skill atrophy from automation complacency: complacency is declining vigilance in a human *supervising* automation; atrophy is degradation of a capability repeatedly bypassed by an AI *replacement*. The distinction matters for intervention design: the former targets monitoring behavior, the latter capability maintenance and deliberate

practice. What offloading theory does not supply is a behavioral formation account: why offloading turns from a strategic choice into a routine, and why it persists once it has.

In AI delegation terms: a maintenance engineer who asks an LLM to generate candidate root-cause hypotheses (rather than producing them from their own diagnostic reasoning) is offloading the very cognitive operation that only repeated practice keeps sharp. Each episode is rational: the AI is faster and the suggestions usually sound. The offloading account explains why the cumulative effect differs from the episodic one, and why the engineer’s bet on the AI’s permanent availability becomes safety-relevant the day the system is down and the diagnosis is urgent.

3.6. Synthesis: cross-field integration and converging empirical evidence

The DHF is not built from scratch: each stage maps onto converging evidence in the literature, including preliminary AI-related findings. Stage 1 (Formation) is the stage trust calibration theory already explains well [21]; Stage 2 (Crystallization) is supported by automation bias evidence showing reliance becomes systematic and resistant to informational intervention [22, 23, 25]; Stage 3 (Divergence) follows from the individual and contextual moderators of habit formation and trust [7, 30]; and Stage 4 (Skill Atrophy) is supported by short-horizon AI evidence [40, 41] and the industrial HRC precedent [17].

Concurrent independent work has reached related conclusions through different routes: self-reported reductions in critical thinking effort among generative AI users [26], a dynamic model in which productivity-skill trade-offs produce an “augmentation trap” under sustained usage [45], and a mixed-method synthesis of AI-induced deskilling in medicine [27]. These accounts document an outcome the DHF explains as a process through habit automaticity.

One circularity concern warrants acknowledgment: the mapping of each field’s contribution in Table 1 draws on the same literature to motivate the DHF and to illustrate its plausibility. This is grounding, not independent validation. Testing the DHF’s specific predictions (particularly the habit-intention gap at Stage 2 and the metacognitive atrophy co-occurrence at Stage 4) requires the longitudinal designs of Section 6.

The following section operationalizes this integrative account into the four-stage Delegation Habit Framework.

Table 1: Four theoretical pillars of the Delegation Habit Framework and their respective contributions.

Field	Key construct	Contribution to DHF	Key gap addressed by DHF
Human-AI trust research	Trust calibration, automation bias	Over-reliance problem; shows reliance decouples from trust over time	No automatization mechanism; no capability outcome
Habit psychology	Habit formation, behavioral automaticity, SRBAI	Formation mechanism; measurement instruments	Not applied to AI delegation
Automation complacency (aviation/HRC)	Complacency, situation awareness degradation	Empirical precedent in physical systems; safety-relevant ancestry	Physical context only; no invisible, persistent habit mechanism
Cognitive offloading theory	Offloading, capability atrophy	Skill degradation mechanism; atrophy vs. complacency	No behavioral formation account

4. The Delegation Habit Framework

4.1. Overview and core propositions

The DHF argues that human reliance on AI systems in repeated work contexts is best understood as a habit-formation process in which deliberate delegation decisions become automatized behavioral routines. The framework makes four core propositions:

1. **Formation proposition:** Repeated AI delegation in stable task contexts produces context-cued behavioral associations that progressively reduce the deliberateness of delegation decisions.
2. **Crystallization proposition:** As delegation routines reach automaticity, they decouple from trust appraisal and persist independently of updated evaluations of AI reliability.
3. **Divergence proposition:** Crystallized delegation habits diverge systematically toward over-delegation for tasks where AI performance is

consistent and contextual cues are strong, and may involve under-delegation where AI exposure is low or task demand for human judgment is explicitly signaled.

4. **Atrophy proposition:** The framework predicts that sustained over-delegation habits will produce measurable degradation in the cognitive capabilities they bypass, reducing independent task performance below pre-adoption levels, a proposition the Skill Retention Assessment Battery (SRAB) protocol (Section 6) is designed to test directly.

These propositions define four progressive stages of the DHF (Figure 1): **Formation**, **Crystallization**, **Divergence**, and **Skill Atrophy** — sequential as a default trajectory, though neither strictly linear nor irreversible (Section 4.6). Although developed in the context of industrial knowledge work, the propositions are domain-general, applying wherever cognitive AI is embedded in repeated professional work — healthcare, education, legal practice, software engineering. Each stage has a distinct behavioral profile, a primary psychological mechanism, and observable indicators.

4.2. Stage 1: Formation

Formation begins with a worker’s first exposures to an AI system in a specific task context. Delegation is deliberate: the worker consciously evaluates whether to use the AI, assesses output quality, and decides whether to accept or override. Trust calibration theory accurately describes this stage, where delegation frequency reflects ongoing trust appraisal as the worker builds a reliability model of the system.

The transition out of Formation begins when the task-context association recurs frequently enough and the AI’s outputs are reliable enough to reinforce the delegation response. Automaticity grows nonlinearly — rapidly at first, then approaching a plateau [30]. The behavioral marker of the transition is delegation without deliberate initiation: the worker begins using the AI before consciously deciding to.

Observable indicators: High variability in delegation decisions across task instances; explicit evaluation behavior; SRBAI scores below the habit formation threshold (a generalized value to be established empirically per context).

4.3. Stage 2: Crystallization

Crystallization is the stage at which delegation behavior reaches its automaticity plateau. The context-response association is now encoded as a

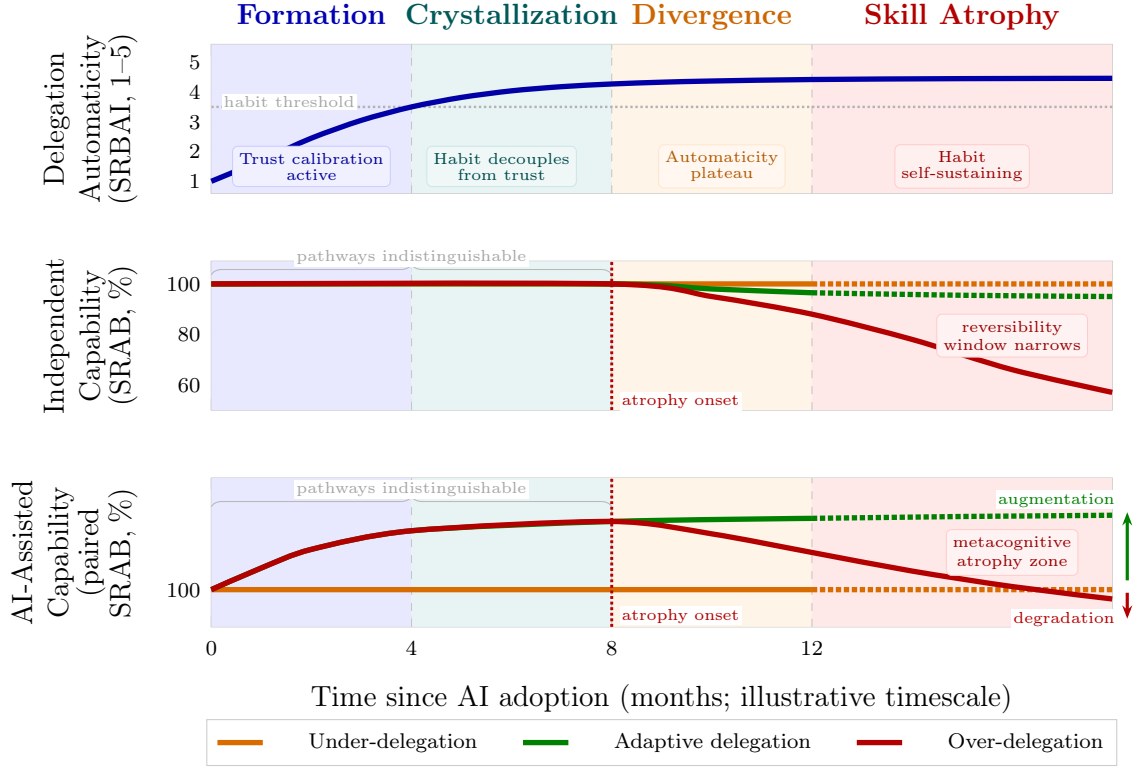


Figure 1: The DHF: a conceptual illustration of predicted trajectories; no curve is based on empirical data, and all numeric axis values are notional; curve shapes and slopes are qualitative estimates grounded in the cross-literature synthesis of Section 3. Both capability panels are snapshots taken *against the worker’s pre-adoption capability profile* (100%). *Top*: delegation automaticity (SRBAI score) grows asymptotically with repeated AI use [30]; the habit threshold, stage boundaries, and timescale are illustrative and vary across tasks, AI systems, and individuals. *Middle*: the worker’s independent capability, measured by the SRAB. Under-delegation preserves the pre-adoption profile; adaptive delegation may dip slightly below it, since less-used pre-adoption capabilities fade while the new, AI-complementary ones being gained are not captured by this baseline; over-delegation erodes it into Stage 4. *Bottom*: AI-assisted capability, on the same baseline, measurable by a paired AI-available administration of the same assessments (Section 6). The two delegating pathways are indistinguishable through Stages 1–2 and diverge at Stage 3: adaptive delegation sustains augmentation, while over-delegation sinks back from what the AI alone provides. In the fourth segment, lines are dotted for under- and adaptive delegation: only over-delegation actually enters Skill Atrophy, while the other pathways merely continue their trajectories. The arrows on the right mark the two directions of movement: upward, augmentation; downward, degradation, which can set in while still above the independent baseline and, if sustained, is the road to substitution. Early SRBAI monitoring identifies at-risk workers before the SRAB reveals them.

behavioral habit: the task-triggering context automatically initiates AI delegation with minimal deliberate mediation, marked by rapid initiation, reduced effort, and resistance to override even when the worker intends to act differently [29].

Decoupling from trust appraisal is the defining feature of Crystallization. Workers here may, when explicitly asked, report moderate or even low trust in the AI, yet their delegation behavior remains consistent and automatic. Trust calibration theory does not readily accommodate this dissociation between expressed trust and behavior, but habit theory anticipates it directly: habitual behaviors are maintained by contextual cues, not motivational states. A worker who intends to verify independently but delegates anyway because the triggering cue is present exhibits classic habit-intention conflict.

Workers at this stage have reorganized their task performance around the AI, so that non-delegation feels cognitively costly. This reorganization is adaptive in stable, high-reliability conditions, but becomes a source of risk when AI reliability degrades, the system is unavailable, or independent judgment is required.

Observable indicators: Low variance in delegation decisions across task instances; significantly reduced verification behavior; SRBAI scores above the habit threshold; explicit accounts of delegation without deliberate initiation (“I just automatically ask it”); weak or absent correlation between expressed trust and delegation behavior.

4.4. Stage 3: Divergence

Divergence describes the systematic differentiation of delegation habits across task types and workers. Cognitive tasks are not equally susceptible to habit formation, and workers do not develop delegation habits at the same rate or degree. The DHF distinguishes two primary trajectories: *over-delegation* and *under-delegation*.

Over-delegation is the dominant pathway for tasks with high contextual consistency, reliable AI outputs, and low explicit demand for independent judgment, conditions under which habit formation operates at full strength and routines persist even when independent performance would be preferable.

Under-delegation characterizes the complementary pathway, arising from limited AI exposure, professional norms of independent performance, or organizational discouragement of AI reliance. It is more than a missed productivity opportunity: in safety contexts, disuse of reliable automation is itself a documented failure mode, as operators who ignore or disable automated

alerts forfeit the protection those systems provide [18]. Where AI reliably outperforms unaided human judgment (e.g., anomaly detection over high-dimensional sensor data), habitual non-delegation produces systematically worse decisions. It also carries a capability cost: the worker never develops the AI-supervision skills (prompt formulation, output verification, failure-mode recognition) that the surrounding socio-technical system increasingly assumes. Under-delegation thus has its own risk profile, and the divergence prediction is symmetric.

Between these poles lies *adaptive delegation*: continued, even frequent, AI use in which verification remains engaged and the underlying capability is periodically exercised. Here delegation stays augmentative: AI-assisted capability rises above the worker’s independent baseline, which under-delegation merely preserves and over-delegation erodes (Figure 1). Even adaptive delegation may reshape the worker’s independent capability profile: less-exercised capabilities may fade while new, AI-complementary ones (task framing, output verification, orchestration) develop. The distinction is between this *transformation* of the capability profile, in which losses are offset by gains, and the net *degradation* of over-delegation, in which little is gained in exchange. Because the SRAB measures capability against the pre-adoption profile, the newly created capabilities do not register on it: adaptive delegation can therefore show a mild decline in the middle panel of Figure 1 even as the worker’s overall repertoire grows.

Over-delegation is nonetheless the DHF’s primary concern, because it leads to Stage 4: delegation so habitual that the outsourced cognitive functions are no longer regularly exercised. The Divergence-to-Atrophy transition is governed by a temporal threshold (how long a capability must be bypassed before measurable degradation begins), likely capability-specific (motor skills may degrade faster than analytical ones, automatized professional skills may resist longer than recently acquired ones), but with a constant mechanism: cognitive functions that are not exercised are not maintained.

Observable indicators: Divergence in SRBAI scores across task classes; behavioral measures of AI usage frequency disaggregated by task type; self-report accounts of task-specific delegation norms; organizational data on AI tool utilization patterns.

4.5. Stage 4: Skill Atrophy

Skill Atrophy is the terminal stage of the over-delegation pathway and the most consequential long-term outcome of crystallized delegation habits.

The mechanism is the cognitive analog of physical deconditioning: a capability that is not regularly exercised gradually loses its functional integrity [13]. The capabilities at risk are the cognitive functions the AI has been substituting: information synthesis, analytical reasoning, judgment under uncertainty, domain-specific pattern recognition, and professional decision-making.

Two qualifications apply. First, atrophy means degradation relative to a counterfactual of maintained independent practice, not necessarily complete loss; the relevant comparison is with a matched worker who continued deliberate practice. Second, partial reversibility through retraining is likely, though the window may narrow as habits crystallize and the work environment increasingly depends on AI-mediated performance (Section 8).

The safety relevance is structurally analogous to aviation’s, though less empirically characterized: a worker who has habitually delegated complex process monitoring faces a capability deficit precisely when the AI fails — under time pressure, during an outage, or on tasks outside the training distribution. At that moment the trust calibration question is secondary; what matters is whether the worker retains enough independent capability to compensate, which the DHF predicts prolonged over-delegation will compromise.

A further dimension is *metacognitive atrophy*: degradation of the worker’s ability to self-assess the capability accurately. [40] and [41] both found AI-assisted skill degradation accompanied by reduced self-assessment accuracy, an illusion of competence in which capability gaps remain invisible (the *intuition rust* of Section 3). Crystallization reinforces this blindspot: the worker who no longer deliberately initiates delegation foregoes the independent performance feedback needed for self-monitoring, and well-calibrated trust alone is insufficient without metacognitive sensitivity [24]. A worker who does not know they have atrophied cannot compensate, which is critical for organizational skill assessment and human-in-the-loop regulatory requirements alike.

Observable indicators: Performance decline on AI-unavailable task instances relative to pre-adoption baseline; divergence between self-assessed and objectively measured capability; reduced accuracy on domain-specific skill assessments administered without AI access; qualitative accounts of discomfort or uncertainty when required to perform AI-delegated tasks independently.

4.6. Framework summary and positioning

Figure 1 represents the DHF schematically. Trust calibration theory gives an accurate account at Stage 1, before automaticity develops. Beyond that

point, habit-based processes dominate, and the relevant outcomes extend beyond reliance decisions to behavioral automaticity, over-delegation frequency, and cognitive skill retention.

Although the stages are presented sequentially, progression is neither strictly linear nor irreversible. Habit theory identifies context change as the most reliable disruptor of established routines [10]: a system redesign, role change, policy shift, or salient AI failure can return a crystallized delegator to deliberate, Stage-1-like evaluation, temporarily or durably. Workers may oscillate around the habit threshold as task contexts vary, and the same worker typically occupies different stages for different task classes at once. The stages are therefore best read as characteristic behavioral regimes (attractor states identifiable by the indicator profiles above) rather than a one-way sequence; the DHF claims only that, absent disruption, the reinforcement structure of reliable embedded AI makes forward movement the default. Reversibility is stage-dependent: behavioral regression is available at every stage, whereas recovery from Stage 4 additionally requires deliberate retraining of the degraded capability, within a window that may narrow as crystallization deepens. Whether post-disruption regression is short-lived (habit decay) or durable (re-calibration), and what magnitude of AI failure triggers it, are empirical questions for the research agenda (Section 7).

Three concurrent, complementary frameworks address related dynamics. The Cognitive Atrophy Paradox model [46] describes a nonlinear co-evolution of cognitive growth and atrophy with critical thresholds, introducing a Cognitive Sustainability Index; the DHF adds the behavioral mechanism explaining why over-delegation develops, and replaces the composite index with independently administrable validated instruments. The Paradox of Augmentation [47] shows via simulation that even idealized reliable AI produces net competency loss, with intermittent-AI designs protective, directly supporting the DHF’s intervention logic. The Augmentation Trap model [45] shows that rational actors over-adopt AI despite known long-run capability decline; the DHF supplies the behavioral mechanism and measurement strategy that make these formal predictions organizationally monitorable.

5. From Physical to Cognitive Human-Machine Teaming: The HRC Bridge

5.1. *The industrial precedent*

Industrial work environments provide the clearest empirical precedent for how sustained automation reshapes human capability. Long before AI agents entered the knowledge worker’s toolkit, automated machinery, robotics, and computer-aided process control generated the phenomena the DHF now extends to cognitive AI. The HRC literature offers a structural precedent rather than a loose analogy: the mechanisms of complacency, capability degradation, and risk amplification documented in physical systems are, the DHF argues, the same mechanisms now operating in the cognitive domain.

The aviation and process control evidence reviewed in Section 3 — degraded manual flying proficiency under sustained automation dependency, operators who lost the plant knowledge needed to manage abnormal conditions [15] — established the pattern in supervisory control settings. HRC has replicated it in shared physical work: the literature raises systematic concern about operator skill retention when robot assistance is withdrawn [17]. The Operator 4.0 concept [35], positioning technology as a human augmentation tool, implicitly acknowledged this risk: augmentation retains its value only if the augmented human retains the underlying capability. Substitution leaves the operator without recourse when the robotic system is unavailable, and for industrial AI deployment this implies that performance gains may be accompanied by gradual erosion of operator capability even under high system reliability.

5.2. *The Operator 5.0 transition and cognitive augmentation*

The Operator 5.0 concept [16], developed within Industry 5.0’s human-centric manufacturing agenda, extends this framework to the cognitive domain. Where Operator 4.0 addressed physical augmentation — exoskeletons, collaborative robotics, sensor-enhanced perception — Operator 5.0 grapples with cognitive augmentation: AI-assisted decision-making, generative AI for design, LLM-mediated knowledge retrieval, agentic AI for process management. Its central tension is the one that animated the HRC skill degradation literature: how to extract the productivity benefits of cognitive augmentation without letting augmentation become substitution.

The DHF locates the crossing point at Crystallization: once delegation is habitual and automatic, the worker outsources cognitive capabilities rather

than augmenting them. The distinction has measurable behavioral correlates (the SRBAI profile, Section 6) and capability consequences (the Stage-4 atrophy trajectory). For organizations invested in the Operator 5.0 agenda, the DHF operationalizes the augmentation-substitution boundary in a way existing frameworks do not.

5.3. Continuities and discontinuities between physical and cognitive teaming

The extension from physical to cognitive human-machine teaming is structural, not metaphorical. Two continuities justify treating the HRC precedent as a generative rather than merely illustrative basis for the DHF.

Task substitution as the common mechanism. In both physical HRC and cognitive AI collaboration, the core dynamic is the progressive substitution of human performance by automated performance within a task class. Whether the task is welding, assembly, or information synthesis, reliable substitution produces the same behavioral consequences: reduced practice, reduced capability maintenance, and increasing dependence on automated performance.

Failure condition criticality. The capability degradation that accumulates under normal operation becomes critical under failure conditions — the robot malfunctions, the AI is unavailable, or the task presents novel features outside the automation’s competence. This is precisely the condition under which human backup capability is most urgently required, yet the automation’s presence has eroded the very capability that would compensate for its absence.

Three discontinuities must be acknowledged. First, cognitive skills are less directly measurable than physical ones: manual dexterity admits direct performance tests, analytical reasoning and professional judgment do not. Second, cognitive AI operates across far more task classes than current collaborative robots [1, 17], which may accelerate habit formation and broaden the capabilities at risk. Third, norms governing cognitive AI delegation are still forming, whereas robotic-assistance norms reflect decades of experience, creating both greater behavioral uncertainty and greater opportunity for normative intervention.

5.4. The DHF in Industrial Practice: An Illustrative Scenario

To make the bridge concrete, consider a maintenance engineer introduced to an AI-assisted predictive maintenance system that ingests equipment sensor data, identifies anomaly patterns, and generates recommendations with

confidence scores. The engineer and their trajectory are hypothetical, but the deployment type is not: AI-assisted predictive maintenance is already operational across advanced manufacturing environments [17], making this the natural setting in which to trace the DHF’s stages.

Stage 1 — Formation. In the first weeks, interaction is deliberate. The engineer scrutinizes each recommendation, cross-references it with their own equipment knowledge, and logs overrides. Trust calibration theory accurately describes this phase: the engineer actively updates a reliability model, calibrates reliance, and retains full situational awareness. SRBAI scores are low; delegation is intentional.

Stage 2 — Crystallization. Over several months, the system proves reliable for the most frequent anomaly classes, and the engineer’s workflow reorganizes around it: opening the dashboard becomes the first response to any alert, before independent inspection, no longer consciously initiated but cued by the notification itself. Verification drops markedly; sensor readings are no longer cross-referenced. SRBAI scores cross the habit threshold; delegation is automatic. Critically, the engineer’s expressed confidence has not changed: what has changed is that behavior is no longer governed by that appraisal.

Stage 3 — Divergence. For routine anomaly classes (the vast majority of alerts), habit crystallization is functionally adaptive: fast, low-effort, accurate. For rare failure modes outside the system’s training distribution, the same automatic response generates risk: the engineer delegates a fault type the system cannot handle, because the cue triggers the habit regardless of the system’s actual competence. The divergence the DHF predicts turns on whether the engineer retains the discriminative awareness to know when not to delegate.

Stage 4 — Skill Atrophy. A plant-wide outage renders the AI unavailable for 72 hours. Confronted with active equipment alerts, the engineer attempts independent diagnosis, but months of habitual delegation have eroded the analytical pattern-recognition skills (reading vibration spectra, interpreting thermal signatures, integrating maintenance history) they would once have exercised routinely. This is not a failure of trust calibration; the engineer trusts the system appropriately. It is a failure of capability maintenance: the HRC analog of the pilot who cannot fly manually when the flight director fails.

The same trajectory operates across industrial knowledge work: a quality engineer under AI-delegated defect detection, or a process engineer whose

reasoning degrades under AI-assisted hazard identification, follows the same arc — the capability at risk differs by domain, the mechanism is identical. The Operator 4.0/5.0 agenda is generative precisely because the substitution dynamic identified in physical systems now operates in the cognitive domain.

5.5. Implications for industrial organizations

The HRC precedent implies a need for deliberate capability maintenance programs in cognitive AI deployment, the analog of aviation’s minimum manual-flight requirements [48], extended to scheduled robot-off practice in collaborative manufacturing cells. The DHF provides both a diagnostic (at what stage are workers for a given AI-delegated task?) and a prescriptive basis for such programs, developed in Section 8.

6. Operationalizing the Delegation Habit Framework

A conceptual framework is useful to industrial organizations only if it enables measuring workforce behavior and capability over time. Two caveats frame what follows. First, the instruments below are largely self-report, and the framework itself predicts that automatized delegation and metacognitive atrophy degrade the introspective access self-report relies on; systematic divergence between self-report and behavior is therefore treated as a signal, not noise, and triangulation with passive, log-derived indicators is integral to the design. Second, every instrument adds a measurement layer, and worker overhead, on top of the AI system and the work itself, so administration frequency and format are themselves critical design parameters. With these caveats in place, this section operationalizes the DHF through three complementary instruments, an integration pathway with industrial information systems, two study designs, and the validity challenges each must address.

6.1. Instrument 1: Adapted Self-Report Behavioral Automaticity Index

The SRBAI [31], based on [32], is a four-item instrument measuring the automaticity component of habit strength for a specified behavior in a specified context. Respondents rate their agreement, from 1 (strongly disagree) to 5 (strongly agree), with four statements describing the target behavior as something they do automatically, without thinking, without having to consciously remember, and without having to make an effort. The score is the mean across items: higher values indicate greater automaticity, that is, a more crystallized delegation habit. The SRBAI has been validated across

many behavioral domains and is sensitive to habit formation trajectories in longitudinal designs [30]. As with any adaptation of a validated instrument, the AI-delegation version requires psychometric verification in context, which the micro-pilot below is designed to provide.

For the DHF, the SRBAI must target AI delegation as the focal behavior, specifying task class and AI system. A sample adapted item: “*When I need to [summarize a technical report / evaluate supplier options / diagnose a process anomaly], using [AI system name] is something I do automatically.*” It should be administered separately per task class, since the DHF predicts task-class-specific trajectories rather than a general AI-use habit.

A longitudinal SRBAI profile lets researchers (a) establish the pre-habit baseline at adoption, (b) track the Formation-to-Crystallization transition as scores cross the habit threshold, (c) identify Divergence by comparing profiles across task classes, and (d) flag over-delegation candidates for targeted skill assessment. Recommended frequency is bi-weekly during Formation and monthly thereafter.

6.2. Instrument 2: Experience Sampling Method Protocol

The Experience Sampling Method (ESM) [49] provides real-time, ecologically valid data on behavior and psychological states in the natural work environment, avoiding retrospective bias and social desirability effects. It is particularly valuable in the Formation stage, when delegation decisions are still deliberate and variable, precisely the variability retrospective surveys average away.

The proposed ESM protocol targets three data streams:

Delegation event capture: Immediately after detected AI tool interactions, participants are prompted (via notification or work-platform integration) with a brief 5-item questionnaire measuring: (1) task type (from a validated taxonomy); (2) whether delegation was consciously initiated; (3) whether and to what degree the AI output was verified; (4) confidence in the output; and (5) experienced effort. This yields a time-stamped behavioral record at the resolution automaticity tracking requires.

Independent task performance capture: At randomized intervals, participants are prompted with a task from their regular AI-delegated task classes and asked to complete it without AI assistance; responses are scored against domain-specific rubrics to yield an ongoing index of independent capability. This is the primary instrument for detecting Skill Atrophy, and it embeds a design tension: independent task probes are themselves deliberate

practice — the very countermeasure Section 8 proposes against atrophy — so a high-frequency schedule would attenuate the effect it is meant to measure. Three mitigations follow. First, probe frequency must be low in absolute terms and relative to the delegated task (roughly one probe per task class per one-to-two weeks, against dozens of delegations). Second, probe density should be randomized *between* participants, so the dose-response relationship between probing and retention can be estimated and the atrophy estimate corrected for practice effects; this correction also protects the comparative predictions, since uncorrected probe-induced practice would compress the very over- versus under-delegator gap the design is built to detect. Third, any residual practice effect biases the *magnitude* estimate conservatively: atrophy observed despite the practice embedded in measurement is a lower bound on what unmonitored workers would experience.

Skill confidence monitoring: Weekly prompts ask participants to rate their confidence in performing each target task class without AI assistance. The longitudinal discrepancy between self-assessed confidence and objectively scored performance measures metacognitive atrophy: the illusion of competence documented by [41].

A complete ESM protocol would run for a minimum of 12 months, with delegation-event sampling densest in the first eight weeks (Formation) and monthly snapshots thereafter; independent task probes follow the deliberately sparse schedule above throughout.

6.3. Instrument 3: Skill Retention Assessment Battery

The SRBAI and ESM provide process measures of habit formation and real-time delegation behavior. Skill Atrophy (Stage 4 of the DHF) requires a dedicated outcome measure: a task-specific assessment of independent capability administered without AI access at multiple timepoints.

Drawing on aviation competency assessment [48, 34], the proposed SRAB consists of: (a) a pre-adoption baseline establishing the capability ceiling; (b) repeated assessments in a standardized AI-unavailable condition, using parallel task forms to limit practice effects, at a cadence that remains an open design parameter: monthly to quarterly during validation, where growth modeling needs timepoints, the sparser 6-month rhythm of aviation-style recurrent checks for routine monitoring; and (c) a meta-cognitive accuracy check in which participants estimate their own performance before receiving feedback. The SRAB thus measures the worker’s *independent* capability; *AI-assisted* capability is measured by a paired administration of

matched tasks with the worker’s normal AI toolchain available, so that the difference between assisted and unassisted scores quantifies the augmentation each pathway delivers (the gap between the bottom and middle panels of Figure 1). The pairing also operationalizes a subtle form of metacognitive atrophy: workers may attribute to themselves capabilities that exist only with the AI’s aid. Where a worker’s self-estimate of unassisted performance tracks their assisted score rather than their actual unassisted one, that attribution error is directly quantified.

Task domains must be defined with domain experts per context (e.g., process anomaly diagnosis, technical document evaluation, risk assessment, engineering decision-making); formats should balance ecological validity with standardizability. The discrepancy between baseline and subsequent assessments, controlling for factors such as organizational learning and role change, is the primary measure of Skill Atrophy.

6.4. *Integration with industrial information systems: toward a delegation audit architecture*

The three instruments above rely substantially on self-report. Industrial environments, however, offer a measurement opportunity that consumer AI contexts lack: work is executed through enterprise systems that already produce event logs: Manufacturing Execution System (MES) and Computerized Maintenance Management System (CMMS) platforms record which work orders, dispositions, and maintenance decisions were taken and when; Supervisory Control and Data Acquisition (SCADA) and alarm-management systems time-stamp operator responses to alerts. The AI side of the trace, by contrast, is not available off the shelf: generic AI assistants expose little or no usage telemetry to the deploying organization. The indicators below therefore presuppose a *deliberately instrumented* AI deployment (for example, AI access routed through an enterprise gateway or platform that logs prompts, responses, and acceptance actions as a design requirement). Under that precondition, joining AI telemetry with existing operational logs turns several DHF indicators into continuously computable behavioral measures [36]; Table 2 sketches candidate operationalizations and their data sources.

These elements compose a *delegation audit architecture* — proposed here as a design template for future deployments, not a description of current practice: a data layer joining AI telemetry with MES/CMMS/SCADA records on task identifiers, an indicator engine computing the measures of Table 2 per worker and task class, and a governance dashboard flagging task classes

Table 2: Candidate log-based operationalizations of DHF behavioral indicators in industrial information systems. The measures are illustrative, not an inventory of current tool capabilities: they presuppose a deliberately instrumented AI deployment, and which of them are obtainable — and at what granularity — depends on the specific systems in place. Self-report instruments (SRBAI, ESM) remain necessary for the constructs that logs cannot capture: experienced automaticity, intention, and confidence.

DHF indicator	Candidate measure	Candidate data source
Delegation frequency per task class	AI invocations per task type and time window	Instrumented AI gateway
Delegation latency (cue-to-delegation)	Time from task trigger (e.g., alert, work order creation) to first AI invocation	SCADA/MES event log joined with AI telemetry
Verification behavior	Coarse proxies, e.g., time between AI response and subsequent action; whether underlying data were accessed in between	AI telemetry; application access logs, where available
Override and modification rate	Changes to AI-proposed dispositions; degree of editing of AI-drafted text	MES/CMMS; document management systems
Independent initiation rate	Share of task instances completed with no AI invocation	MES/CMMS records cross-referenced with AI telemetry
Atrophy signal (Stage 4)	Performance and completion time on tasks executed during AI unavailability windows	MES/CMMS records during outages or scheduled AI-free operation

drifting toward Crystallization (rising delegation frequency, latency compression, falling verification and independent initiation) for the organizational responses of Section 8. Realizing this is itself a system design challenge (instrumentation, identifier management, log quality), and thus an item for this community’s research agenda rather than its inventory. Three constraints are integral. First, worker monitoring of this kind falls within data protection and labor law (GDPR and, in many European settings, works council co-determination); indicators should be computed at the least granular level that supports the governance purpose, aggregated across workers wherever individual identification is not strictly required. Second, log-derived measures are proxies: dwell time indicates the *opportunity* to verify rather than verification itself, and requires periodic validation against ESM self-report. Third, the architecture measures behavior, not capability: logs identify *who should be assessed*, while capability is established only by the SRAB. Together they give organizations monitoring coverage no self-report protocol can match, and give the DHF a measurement pathway native to industrial information systems.

Beyond self-report and system logs, a third measurement modality deserves brief note: direct physiological monitoring. Non-invasive wearables (capturing, e.g., heart rate variability, electrodermal activity, or gaze) and lightweight electroencephalography headsets can index cognitive load, vigilance, and engagement in real time [50, 51], offering a window onto constructs that neither logs nor self-report capture well: the declining cognitive effort of delegation as automaticity grows, a candidate physiological signature of the Formation-to-Crystallization transition, and the attentional disengagement that accompanies verification decline. Field validity, comfort, worker acceptance, and stricter data protection constraints remain open challenges; we flag this modality as a complement to the instruments above, not a substitute.

6.5. Study designs for DHF validation

Design 1: Longitudinal ESM micro-pilot ($N = 20\text{--}30$, 8 weeks). The recommended first step tracks SRBAI and ESM data across the early Formation stage for knowledge workers beginning use of a specific AI system for a specific task class. The primary outcome is whether SRBAI scores show the asymptotic growth trajectory predicted by habit theory [30]; secondary outcomes include the correlation between SRBAI and ESM-measured deliberateness, and early signals of independent capability change. The micro-

pilot is scoped for feasibility, protocol refinement, and effect size estimation rather than confirmatory testing: its intensive repeated measurement supports within-person growth modeling, but between-person inferences at this sample size are exploratory by design.

Design 2: Full longitudinal study ($N \geq 60$, 12 months). The primary validation study tracks participants from initial AI adoption through Crystallization and Divergence, with monthly-to-quarterly skill retention assessments from baseline to M12. Key predictions: (a) SRBAI scores follow an asymptotic growth curve; (b) habit strength outpredicts trust: SRBAI scores measured at M3 explain more of the variance in delegation frequency at M6 and M12 than trust scales also measured at M3; (c) high-SRBAI participants on capability-critical tasks show significantly larger SRAB declines at M12 than matched low-SRBAI participants. The $N \geq 60$ figure is indicative; the definitive sample size must come from a simulation-based power analysis for multilevel growth models, parameterized with micro-pilot variance and effect size estimates. A 12-month design is a plausible minimum for detecting Skill Atrophy, given that capability degradation is slow-moving; the actual detection window is itself an empirical question (Section 7).

6.6. *Validity considerations*

Four validity threats require management. **Selection effects:** over-delegators may differ on individual characteristics (need for cognition, AI anxiety) that also predict skill maintenance; within-participant designs and pre-adoption baselines partially mitigate this. **Ecological confounds:** organizational and technological change over the study period must be documented to avoid confounding capability trajectories. **Demand characteristics:** covert measurement via system logs, where ethically permissible, provides a behavioral check on self-report. **Pre-existing AI use:** most knowledge workers already use AI at study entry; participants at varying DHF stages can be analysed as a quasi-longitudinal cohort, with retrospective baselines from early-adoption task classes or organizational records of tool introduction.

7. A Research Agenda for the Delegation Habit Framework

The DHF generates a structured empirical agenda organized across the four stages of the model. The eight research questions (RQs) below are prioritized in approximate order of logical dependency: earlier questions establish

conditions for later ones, and the research agenda is designed to be pursued sequentially by a longitudinal program, or selectively by independent investigators.

7.1. Formation stage

RQ1: Do AI delegation behaviors follow the asymptotic habit formation trajectory? The foundational claim is that AI delegation automaticity (SRBAI) follows the asymptotic growth curve documented by [30]. Testable in the 8-week micro-pilot (Design 1, Section 6); null findings would indicate that the time scales or task characteristics of AI use require modifying the habit formation model.

RQ2: What task and system characteristics moderate the rate of delegation habit formation? Habit formation speed is moderated by contextual consistency, outcome reliability, and effort [30, 52]. The candidate AI-delegation moderators are task contextual consistency, AI output reliability, and task complexity, the factors the predictive maintenance scenario (Section 5) illustrates as governing the Formation-to-Crystallization transition rate.

7.2. Crystallization stage

RQ3: Does habit strength (SRBAI) predict AI delegation behavior beyond trust measures? Administering both SRBAI and a validated trust scale [7] longitudinally, the DHF predicts that SRBAI will explain more behavioral variance after the habit formation period, with the advantage growing over time: direct evidence that habit plays a stronger explanatory role than trust recalibration in sustained AI use, complementing trust calibration’s account of Formation.

RQ4: Does the habit-intention gap emerge in AI delegation? The gap between expressed intention and actual behavior in favor of the established habit [29] is a canonical marker of crystallized habit. Documenting its prevalence in industrial knowledge work would validate the Crystallization stage and identify workers who have lost volitional control over their delegation behavior.

7.3. Divergence stage

RQ5: What individual, task, and organizational factors predict over-delegation versus under-delegation divergence? The DHF predicts systematic divergence at Stage 3, but which workers take which

pathway, for which tasks, remains open. Candidate factors include need for cognition, AI-specific anxiety, professional identity threat, and prior automation experience (individual); AI reliability, task criticality, and norms of independent performance (task); and governance policies, team norms, and managerial signaling (organizational). A survey of knowledge workers at varying adoption stages, combined with behavioral delegation measures, is the recommended design.

RQ6: Do over-delegators and under-delegators show divergent cognitive capability trajectories at 12 months? This is the pivotal question connecting Stage 3 (Divergence) to Stage 4 (Skill Atrophy). The prediction is that participants classified as over-delegators at Stage 3 (high SRBAI on capability-critical tasks, high delegation frequency, low verification behavior) will show significantly larger SRAB score declines at T12 than participants classified as under-delegators or adaptive delegators. Establishing this relationship empirically would provide the first direct evidence linking behavioral habit measures to cognitive capability outcomes in human-AI collaboration research.

7.4. Skill Atrophy stage

RQ7: What is the time course and reversibility of skill atrophy following over-delegation habit crystallization? If Skill Atrophy is established, the next questions concern its trajectory and reversibility: does atrophy accumulate linearly or plateau, and is it fully reversible through deliberate retraining? The answers govern how long organizations can allow over-delegation habits to persist before irreversible atrophy risk becomes substantial, and what retraining investment restores capability. A parallel question concerns individual agency: why should a worker invest in skill maintenance when AI performs the task better and faster? Understanding motivation to retrain — intrinsic (professional identity, autonomy, resilience) or extrinsic (incentives, appraisal, regulatory mandate) — is essential for designing retraining programs workers will engage with rather than minimally comply with.

RQ8: Does metacognitive atrophy (the illusion of competence) co-occur with skill atrophy, and does it mediate the safety consequences of over-delegation? [41] provided initial evidence that AI-assisted skill degradation is accompanied by inaccurate self-assessment. If the co-occurrence is reliable, it has major governance implications: an organization relying on workers to self-identify capability gaps may systematically fail to

detect those most at risk. The question is addressable within the SRAB protocol, by comparing self-assessed with objectively scored performance across the atrophy trajectory.

7.5. Cross-domain generalization

The eight RQs are specified for industrial knowledge work, but the DHF’s propositions are domain-general. Cross-domain replication in healthcare, legal practice, software engineering, education and scientific research would establish generalizability and identify domain-specific moderators, and the stakes are rising fastest in the latter domains: AI-assisted diagnostics creates atrophy risk for clinical reasoning already documented in radiology [27]; AI code completion may erode independent implementation reasoning, a risk already visible among novice programmers in computing education [53]; and AI-assisted science risks producing illusions of understanding in researchers themselves [54]. Of particular theoretical interest is accountability pressure: settings where errors carry high personal or legal consequences may show different Divergence patterns, with accountability functioning as a habit interruption cue even after Crystallization.

8. Discussion

8.1. Implications for system and interaction design

The DHF reframes the design question from *whether* knowledge workers should use AI to *how delegation decisions remain conscious over time*. Augmentation is possible only while delegation remains in the effortful-choice regime; once crystallized, AI executes cognition on the worker’s behalf. Design should therefore *protect deliberation, not ration access*. Figure 2 renders this prescription as a map. Each worker and task class occupies a position in an engagement–delegation plane, with both coordinates measurable (delegation from logs; engagement from inverse SRBAI automaticity and verification indicators). To our knowledge, it is the first map of AI delegation against the worker’s cognitive engagement, complementing outcome-based reliance taxonomies [55] and skill-based delegation phase planes [56]. Augmentation is available only in the upper-right region; habit formation is a downward drift across the threshold into use that continues unchosen and unchecked. Interventions therefore act vertically, restoring engagement, not horizontally, rationing access.

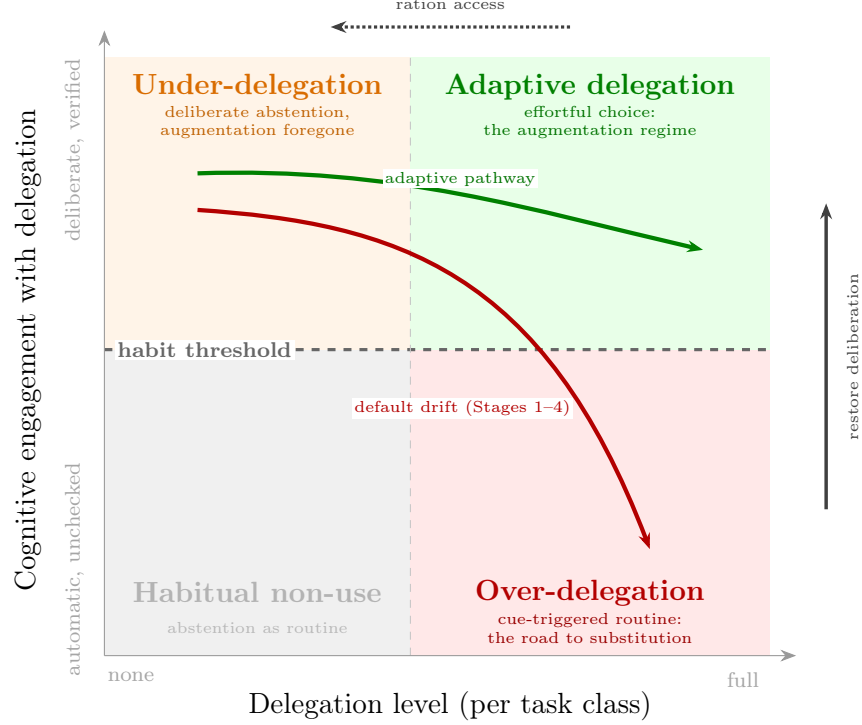


Figure 2: The delegation map: the DHF’s state space for a worker and task class. The horizontal axis is the level of delegation (measurable from system logs); the vertical axis is the worker’s cognitive engagement with delegation, a composite of deliberate initiation and verified acceptance (components the framework predicts decline together), with the habit threshold separating engaged use from cue-triggered routine. Four regimes follow, with illustrative boundaries: *adaptive delegation*, extensive but consciously initiated use, the only region where augmentation is available; *over-delegation*, use that continues without conscious choice, the road to substitution; *under-delegation*, deliberate abstention that foregoes augmentation; and *habitual non-use*, abstention that has itself become routine. The red arrow traces the default drift of Stages 1–4 across the threshold; the green arrow, the adaptive pathway that countermeasures aim to sustain. Both arrows generalize predicted trends rather than precise paths: individual trajectories differ in slope and pace, may reverse after context change, and the same worker can follow different paths for different task classes (Section 4). The two outer arrows are the corrective mechanisms for these drifts. The solid vertical arrow is the intervention the framework prescribes: restoring deliberateness while preserving delegation. The dotted horizontal arrow, rationing access, is the discouraged alternative: it sacrifices augmentation without rebuilding engagement. Plotted from live indicators, the map turns the measurements of Section 6 into an organizational diagnostic: position locates a worker; drift toward the lower right is the alarm.

The dominant design philosophy for AI interfaces has been frictionless integration: making access as fast and effortless as possible. From a habit formation perspective, frictionless design is precisely the condition that maximizes automaticity growth [10]: every choice that removes deliberate initiation (contextual auto-suggestion, one-click access, proactive output generation) accelerates the Formation-to-Crystallization transition and raises over-delegation risk for capability-critical tasks.

A DHF-informed design philosophy inverts this logic for capability-critical tasks where atrophy poses organizational or safety risk: there, deliberate friction is not a usability failure but a protective mechanism. Interventions derived from the DHF include: (a) *mandatory delegation declaration*, requiring users to actively confirm delegation before AI assistance is provided; (b) *periodic AI withdrawal*, removing assistance for targeted task classes on a scheduled basis; (c) *verification prompts*, requiring users to evaluate outputs before accepting them; and (d) *capability scaffolding modes*, in which the AI progressively reduces output detail to require increasing independent contribution.

These interventions are not speculative; analogues exist in aviation training (minimum manual flying requirements [48]), medicine, where AI-induced deskilling has prompted analogous safeguards [27], and the capability maintenance concerns raised in HRC research [17]. The DHF provides the behavioral mechanism that explains why these interventions work and the measurement instruments that can evaluate their effectiveness.

8.2. Implications for organizations

Standard AI governance focuses on system reliability, explainability, and fairness, properties of the AI system [57]. The DHF identifies a complementary requirement that existing frameworks address only partially: monitoring the behavioral and capability consequences of AI use on the workforce. Paradoxically, organizations deploying the most reliable AI systems may create the conditions for Divergence and Skill Atrophy at the greatest scale: their systems sustain the reinforcement crystallization requires.

Organizational responses suggested by the DHF include: (a) *AI delegation audits*, combining adapted SRBAI measures with the log-derived behavioral indicators of the delegation audit architecture (Section 6.4) to assess habit strength for capability-critical task classes; (b) *skill retention assessments*, administering the SRAB periodically without AI access; (c) *deliberate practice requirements*, building AI-free task performance into role expectations

and appraisal; and (d) *role definition review*, distinguishing tasks appropriate for automatized delegation from those requiring ongoing independent practice. Plotted across a workforce, these indicators render the delegation map (Figure 2) as a heat map, flagging workers and task classes drifting toward over-delegation.

Gartner forecasts that, through 2026, GenAI-driven atrophy of critical-thinking skills will push 50% of global organizations to require AI-free skills assessments [58], precisely the audit the DHF operationalizes through the SRAB. The productivity-capability trade-off has aviation precedent: minimum manual-flight requirements trade a productivity cost for retained capability.

Beyond individual organizations, workforce-level skill atrophy raises *social sustainability* concerns: a generation of workers who never developed the independent capabilities AI now performs represents a systemic fragility, acute during outages, regulatory restrictions, or tasks the AI cannot handle. A workforce capable of co-existing with AI rather than merely depending on it is a societal resilience asset [16].

8.3. Implications for regulatory design

The EU AI Act [57] introduces a risk-tiered regulatory framework for AI systems deployed in European markets, with stringent requirements for high-risk applications in sectors including manufacturing, healthcare, transport, and critical infrastructure. Its human oversight mandate is set out in Article 14: high-risk AI systems must be designed so that natural persons can effectively oversee them, and Article 14(4) enumerates the capabilities the oversight measures must enable — including, at point (b), remaining aware of the possible tendency to automatically rely or over-rely on the system’s outputs (automation bias), the very phenomenon this paper traces to its behavioral roots.

The DHF reveals a critical gap in this mandate as formulated. Article 14 locates compliance in system design and provider obligations: does the system make effective oversight *possible*? It does not assess the human side over time: are the persons assigned to oversight still *capable* of exercising it after months of habitual delegation? An organization deploying a technically compliant high-risk system while its operators’ oversight capabilities atrophy through over-delegation habits may be formally compliant yet substantively failing the requirement: the mechanisms exist, but the humans who must operate them no longer can.

This suggests a regulatory extension: human-in-the-loop requirements for high-risk AI systems should include capability maintenance provisions analogous to those in aviation and medical device regulation: periodic competency assessments, minimum AI-unassisted task performance, and documented independent capability for operators of safety-relevant systems. The DHF provides the conceptual and measurement basis for such provisions.

8.4. Limitations

The DHF is a conceptual framework without direct empirical validation yet. Its propositions are grounded in four established literatures and in the documented precedents of physical human-machine teaming, but its specific predictions for cognitive AI delegation (habit formation trajectories, the habit-trust divergence, the atrophy time course) remain to be tested through the operationalization (Section 6) and research agenda (Section 7).

The immediate next empirical step is a living-lab pilot in which the research team serves as the first participant cohort (appropriate given that researchers, engineers, and educators are among the most intensive users of cognitive AI). The planned 8-week study applies the SRBAI and ESM protocol with pre-study SRAB baselines; participants already using AI will be profiled at intake for retrospective baseline reconstruction. External validation with industrial partners is planned subsequently.

Additionally, the DHF is specified at a level of abstraction requiring domain-specific elaboration: the task taxonomy, SRAB domains, and intervention designs must be developed with domain experts per setting.

Finally, the DHF focuses on the negative consequences of habitualized delegation and does not account for the efficiency and cognitive-load benefits that well-calibrated delegation habits provide; a full account will need to balance this risk-oriented analysis against the conditions under which habitualized delegation is a productive adaptation.

9. Conclusion

As agentic AI embeds itself in industrial knowledge work, trust calibration theory (designed for episodic, expert-mediated automation) insufficiently accounts for the behavioral dynamics of sustained use. This paper has proposed the Delegation Habit Framework as a complementary account, grounded in converging evidence from habit psychology, human-AI trust research, automation complacency, and cognitive offloading theory.

The DHF’s central argument is that AI adoption is a habit-formation problem as much as a trust problem, and that over-delegation habits can measurably degrade the cognitive capabilities they replace. The dynamic is a recognizable consequence of sustained, reliable automation, documented across decades of aviation and human-robot collaboration; cognitive human-machine teaming extends it to a domain where the capabilities at stake are analytical reasoning, professional judgment, and domain expertise, and where degradation is harder to detect and remediate. Organizations deploying cognitive AI, regulators writing oversight requirements, and designers encoding assumptions into AI interfaces all stand to benefit from a framework that makes this risk visible, the mechanism explicit, and the measurement tractable. The DHF provides that framework.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

Ironically, a paper about AI-induced skill atrophy that was written without AI assistance would have been more poetically consistent — but less honest, and arguably less interesting. Claude Opus 4.8 and Fable 5 (Anthropic) were used throughout the writing process for cross-literature discovery and referencing, conceptual brainstorming, prose and figures drafting, and manuscript formatting. Research is, after all, a paradigmatic example of the knowledge work this paper addresses. However, every AI-generated output was scrutinized, challenged, and revised by several — not yet atrophied

— human brains, whose analytical reasoning, professional judgment, and domain expertise remain, we believe, intact. The authors take full responsibility for all intellectual content, all errors, and the decision to delegate deliberately rather than habitually. Deliberate delegation augments capability; the paper you have just read is our attempt to practice what it theorizes.

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